

Can someone explain to me how AI is directly bad for the environment?

Posted by u/Ok-Examination4225 • 0 points • 67 comments

Outside of, "it uses a lot of energy, so that makes it bad", because that's not really a problem because energy can be green.

Comments

u/Worth_Inside421 • 0 points • 2025-08-13 04:26 UTC

Hello! contrary to what many people believe in these replies, AI is in fact very dangerous for the environment. It adds to climate change crisis especially. For one it contributes heavily to the greenhouse gas emissions due to the large scale energy it uses. Second, it uses an immense amount of fossil fuels due to the large amount of electricity used. And one of the biggest reasons being the immense amount of water used for cooling down data centers where the ai operates. While it may not seem like a lot of water, .3 ml of water are used per prompt /query (coming from CEO of OpenAI. Keep this in mind because scientists and researchers are finding that number to be closer to 10ml). With that stat being used,

1 billion AI queries per day (a conservative estimate)

× 0.3 mL per prompt

= around 85,000 gallons of water used daily

That's equivalent to the water usage of roughly 1,000 American households, based on EPA benchmarks (averaging ~82 gallons per home, per day).

u/InterestFancy8668 • 2 points • 2025-08-06 21:02 UTC

TikTok uses more water than ChatGPT btw

u/FDHed • 1 points • 2025-07-04 16:47 UTC

"Energy can be green" is doing a lot of heavy lifting here lol

u/Ok-Examination4225 • 1 points • 2025-07-04 22:35 UTC

Energy will realistically never be 100% green and that's not really the point. I was just interested why is everyone itching about it. Oh no it consumes power. Dude like, Everything, consumes power.

u/jferments • 2 points • 2025-07-04 16:27 UTC

It's a bad argument made by misinformed anti-AI zealots, and if you actually dig into the data on energy usage, you'll find there is nothing there. The arguments against AI are basically arguments against modern computing (the large majority of data center energy consumption is NOT for AI). Sure if you want to shut down data centers and not have the modern internet, there are some environmental

arguments that could be made, but these are not unique to AI.

Also, there are things like industrial meat production, concrete manufacturing, and war (e.g. the DOD is the world's largest polluter, by far) that use vastly more resources and cause orders of magnitude more harm than AI.

u/Corren_64 • 2 points • 2025-07-04 15:40 UTC

Creating 1000 pictures takes between 1 to 4 kWh. So naturally it is way better to have one artist work on their 500W computer for 20 hours to get one picture.

u/Artholos • 1 points • 2025-07-04 14:16 UTC

The electricity cost for me to generate one 1080x1080 image on my 3090 Ti is about 1/60th of one cent USD, iirc. I haven't even calculated how much is saved from the crazy speed increase of my 5080...

I'd have to be cranking out images 24/7 week all month to work up a bill.

u/Unlucky-Scallion1289 • 1 points • 2025-07-04 12:42 UTC

Plenty have already covered the energy costs. So I'll some additional context.

Researchers are shifting towards AI instead of supercomputers in some instances. The initial cost to train AI is very high but following uses doesn't cost much at all. A single LLM inference, like a ChatGPT reply, costs practically zero, equivalent to mere milligrams of CO2.

For example, training GPT-4 costs a CO2 equivalent to 8000 - 20,000 metric tons. A single supercomputer task over 1 week on 1,000 nodes costs a CO2 equivalent to 120 - 400 metric tons.

So after approximately 50 - 67 supercomputer uses the AI models become more efficient. That's a little over a year of compute power. The question then is whether or not it's worth it to spend 1 year of compute power in order to increase future efficiency.

Of course, AI isn't suitable for every task that supercomputers can do. So it heavily depends on the research being done. But it's fascinating how AI can push the edge of science.

u/Maleficent_Sir_7562 • 1 points • 2025-07-04 12:24 UTC

It's actually not bad at all

Studies before have shown an incorrect figure of about 500 millimeters per 10-20 ChatGPT prompts back in 2023.

This exaggerated over a thousand times on the real number. OpenAI themselves released the numbers, and the actual water usage of one prompt is 0.000085 gallons, or one fifteenth of a teaspoon. In terms of power, if I remember it's about 0.6 watt hours, enough to power a light bulb for a minute or so or turn on a microwave/oven for a few seconds.

Even at hundreds of millions of prompts, this goes only for about ten thousands of gallons per day.

And one hamburger is 632 gallons. Double beef ones can be over a thousand. McDonald's sells 6 million of these a day.

That means the entire ChatGPT usage global water usage with a few hundred million prompts is quite literally equivalent to a few dozen hamburgers a day.

Even if you count training, which is the only expensive part that does use a lot of water or power... it doesn't match to 1% of daily hamburger sales, and that's just McDonald's only.

With one day of hamburgers in McDonald's, you could train GPT-4 like 25000 times.

Seeing this, I would say it's pretty damn efficient compared to the beef industry. Especially with the fact that many jobs done by a human actually use more energy than ai doing it, cuz you'll have to leave your computer on or operate on it for longer, and etc.

And even though it's like this, big companies like Microsoft are already making advances and built nuclear power plants for ai, and nuclear power is really efficient and reliable.